

Non-Translatable Technical Glossary

Computer Science & Software Engineering

NPTEL Course Translation Pipeline · Do-Not-Translate (DNT) Reference

This glossary defines all technical terms that must remain in their original English form during translation into Hindi, Telugu, Tamil, and Kannada.

Version 1.0 · Alphabetically Grouped A–Z · Beginner-Friendly Definitions

133

Total Terms

25

Letters Covered

10

Categories

4

Target Languages

HOW TO USE THIS GLOSSARY

This document is the authoritative Do-Not-Translate (DNT) reference for the NPTEL course translation pipeline. Every term listed here must appear unchanged — in its exact English form — in any translated output, regardless of target language. These terms must never be translated into Hindi, Telugu, Tamil, Kannada, or any other language.

- Feed this document to your LLM as a system-level instruction or RAG context chunk.
- Each entry can be independently indexed as a vector in ChromaDB for retrieval.
- The Category badge (e.g., DB, OOP, ALGO) can be stored as metadata for filtered lookup.
- Terms that appear as acronyms (SQL, API, OOP) must retain that exact capitalisation.
- If a listed term appears mid-sentence in source content, it must remain in English in the translation.

CATEGORY LEGEND

Object-Oriented	Database	CS Concept	Algorithm	Network
DevOps / Cloud	Machine Learning	Security	Programming	Systems

ALPHABET INDEX

A	B	C	D	E	F	G	H	I	J	K	L	M
10	8	14	13	6	5	3	4	9	2	1	4	7
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
2	6	7	3	7	9	5	2	3	1	1	—	1

A

10 terms

Abstract Class

Object-Oriented

A class in OOP that cannot be instantiated directly and is meant to be subclassed, providing a common interface for derived classes.

Abstraction

Object-Oriented

An OOP principle that hides complex implementation details and exposes only the necessary interface to the user.

ACID

Database

A set of properties (Atomicity, Consistency, Isolation, Durability) that guarantee reliable database transactions.

Algorithm

CS Concept

A finite, step-by-step procedure for solving a problem or accomplishing a computation task.

API

CS Concept

Application Programming Interface — a defined set of rules that allows different software applications to communicate with each other.

Array

CS Concept

A data structure that stores a fixed-size, ordered collection of elements of the same data type.

Asynchronous

Programming

A programming model where operations execute independently without blocking the main program flow, using callbacks, promises, or async/await.

Atomicity

Database

A database transaction property ensuring all operations in a transaction either fully complete or are fully rolled back.

Authentication

Security

The process of verifying the identity of a user, system, or application before granting access.

Authorization

Security

The process of determining what actions or resources an authenticated user is permitted to access.

B

8 terms

B-Tree

Database

A self-balancing tree data structure that maintains sorted data and allows efficient insertion, deletion, and search operations, widely used in databases.

Big O Notation

Algorithm

A mathematical notation used to describe the upper bound of an algorithm's time or space complexity in the worst case.

Binary Tree

CS Concept

A tree data structure in which each node has at most two children, referred to as the left child and the right child.

Buffer

Systems

A temporary memory area used to hold data while it is being transferred between two locations or processes.

Bandwidth

Network

The maximum data transfer rate of a network connection, typically measured in bits per second (bps or Mbps).

Binary Search

Algorithm

An efficient search algorithm that finds a target value in a sorted array by repeatedly halving the search interval.

Boolean

Programming

A primitive data type with only two possible values: true or false, used in logical conditions and control flow.

Bug

CS Concept

An error, flaw, or unintended behavior in a software program that causes it to produce incorrect or unexpected results.

C

14 terms

Cache

Systems

A high-speed storage layer that stores copies of frequently accessed data to reduce retrieval time from slower storage.

Candidate Key

Database

An attribute or set of attributes in a database table that can uniquely identify each row and is eligible to be the primary key.

Client

Network

A computer or software application that requests services or resources from a server in a client-server architecture.

Cluster

DevOps / Cloud

A group of interconnected computers or nodes that work together as a single system to provide high availability and scalability.

Composite Key

Database

A primary key made up of two or more columns in a database table that together uniquely identify a row.

Consistency

Database

A database property ensuring that any transaction brings the database from one valid state to another valid state.

Container

DevOps / Cloud

A lightweight, portable, and isolated software unit that packages code and its dependencies together for consistent execution across environments.

Callback

Programming

A function passed as an argument to another function, to be executed after a certain event or operation completes.

Class

Object-Oriented

A blueprint or template in OOP that defines the attributes (data) and methods (behavior) of objects created from it.

Cloud Computing

DevOps / Cloud

The delivery of computing services — servers, storage, databases, networking, software — over the internet on a pay-as-you-go basis.

Compiler

CS Concept

A program that translates source code written in a high-level programming language into machine code or bytecode.

Concurrency

Systems

The ability of a system to execute multiple tasks or processes simultaneously or in overlapping time periods.

Constructor

Object-Oriented

A special method in a class that is automatically called when an object is created, used to initialize the object's attributes.

CRUD

Database

Create, Read, Update, Delete — the four fundamental operations performed on data in a database or persistent storage system.

D

13 terms

Data Definition Language

Database

DDL — a subset of SQL used to define and modify database structure, including CREATE, ALTER, DROP, and TRUNCATE commands.

Data Structure

CS Concept

A specialized format for organizing, storing, and managing data in memory so it can be accessed and modified efficiently.

Database

Database

A structured collection of data organized for efficient storage, retrieval, and management, typically controlled by a DBMS.

Deadlock

Systems

A situation in a multi-process system where two or more processes are blocked indefinitely, each waiting for the other to release a resource.

Denormalization

Database

The intentional process of adding redundancy back into a normalized database to improve query read performance.

Deployment

DevOps / Cloud

The process of releasing and installing a software application or update into a target environment for end-user use.

Dynamic Programming

Algorithm

An algorithmic technique that solves complex problems by breaking them into overlapping subproblems and storing results to avoid redundant computation.

Data Manipulation Language

Database

DML — a subset of SQL used to retrieve and modify data, including SELECT, INSERT, UPDATE, and DELETE commands.

Data Type

Programming

A classification that specifies the type of value a variable can hold, such as integer, float, string, or boolean.

Database Management System

Database

DBMS — software that provides tools to create, manage, and interact with databases, handling storage, retrieval, and security.

Debugging

CS Concept

The systematic process of finding, identifying, and fixing errors or bugs in a software program.

Dependency

CS Concept

In databases, a functional relationship between attributes; in software, a module or library required by an application to run.

Destructor

Object-Oriented

A special method in a class that is automatically called when an object is destroyed, used to release resources held by the object.

E

6 terms

Encapsulation

Object-Oriented

An OOP principle that bundles data and methods together inside a class and restricts direct access to the internal state.

Encryption

Security

The process of converting plaintext data into an unreadable ciphertext format using a cryptographic algorithm and key.

Entity

Database

In a database context, a distinct real-world object or concept — such as a Student or Product — that has attributes and is stored as a table.

Entity-Relationship Diagram

Database

ER Diagram — a visual model that represents entities, their attributes, and the relationships between them in a database design.

Exception

Programming

An abnormal condition or error event that disrupts the normal flow of a program's execution and must be handled explicitly.

Exception Handling

Programming

The mechanism in programming for responding to runtime errors using constructs like try, catch, and finally blocks.

F

5 terms

File System

Systems

The method an operating system uses to organize, store, name, and retrieve files on storage media.

Framework

CS Concept

A pre-built, reusable software platform that provides a foundation and standard structure for developing applications.

Functional Dependency

Database

A relationship in a database where the value of one attribute uniquely determines the value of another attribute.

Foreign Key

Database

A column or set of columns in a database table that references the primary key of another table, enforcing referential integrity.

Full Outer Join

Database

A SQL join operation that returns all rows from both tables, filling in NULL for non-matching rows on either side.

G

3 terms

Garbage Collection

Systems

An automatic memory management process that identifies and frees memory occupied by objects no longer referenced by the program.

Greedy Algorithm

Algorithm

An algorithmic approach that makes the locally optimal choice at each step, aiming to find a globally optimal solution.

Graph

CS Concept

A non-linear data structure consisting of nodes (vertices) and edges (connections) used to model relationships and networks.

H

4 terms

Hashing

CS Concept

The process of mapping data of arbitrary size to a fixed-size value using a hash function, used for fast data lookup and integrity checks.

Heap

CS Concept

A specialized tree-based data structure where the parent node is always greater than (max-heap) or less than (min-heap) its children.

HTTP

Network

Hypertext Transfer Protocol — the foundation of data communication on the web, defining how clients and servers exchange requests and responses.

HTTPS

Network

HTTP Secure — an encrypted version of HTTP that uses SSL/TLS to protect data in transit between client and server.

9 terms

Index

Database

A database object that improves the speed of data retrieval operations by creating a separate, optimized lookup structure.

Inner Join

Database

A SQL join that returns only the rows where there is a matching value in both tables being joined.

Integer

Programming

A whole-number data type that stores numeric values without a fractional component, typically in ranges like int (32-bit) or long (64-bit).

Interpreter

CS Concept

A program that executes source code directly, line by line, without prior compilation into machine code.

Isolation

Database

A database transaction property ensuring that concurrent transactions execute independently without interfering with each other.

Inheritance

Object-Oriented

An OOP principle where a child class acquires the properties and methods of a parent class, enabling code reuse and hierarchical classification.

Instance

Object-Oriented

A specific realization of a class — an individual object created from the class blueprint with its own unique attribute values.

Interface

Object-Oriented

An OOP contract that defines a set of method signatures a class must implement, without providing any implementation itself.

IP Address

Network

Internet Protocol Address — a unique numerical label assigned to each device on a network for identification and location addressing.

J

2 terms

JOIN

Database

A SQL operation that combines rows from two or more tables based on a related column to retrieve unified result sets.

JSON

CS Concept

JavaScript Object Notation — a lightweight, human-readable data interchange format using key-value pairs, widely used in APIs and configuration.

K

1 term

Kernel

Systems

The core component of an operating system that manages communication between hardware and software, handling memory, processes, and I/O.

L

4 terms

Latency

Network

The time delay between initiating a request and receiving the first byte of a response, typically measured in milliseconds.

Left Outer Join

Database

A SQL join that returns all rows from the left table and matching rows from the right table, with NULL for non-matches.

Linked List

CS Concept

A linear data structure where each element (node) contains data and a pointer to the next node, forming a chain.

Load Balancer

DevOps / Cloud

A system that distributes incoming network traffic across multiple servers to ensure no single server is overwhelmed.

M

7 terms

Machine Learning

Machine Learning

A subset of AI where systems learn patterns from data and improve their performance on tasks without being explicitly programmed.

Metadata

CS Concept

Data that describes other data — for example, a file's creation date, size, and author are metadata about the file itself.

Microservice

DevOps / Cloud

An architectural style where an application is built as a collection of small, independently deployable services, each responsible for a single function.

Multithreading

Systems

A programming technique where multiple threads of execution run concurrently within a single process to improve performance.

Memory Management

Systems

The process of allocating, tracking, and releasing computer memory during a program's execution to prevent leaks and optimize performance.

Method

Object-Oriented

A function defined inside a class that describes the behaviors or actions that objects of that class can perform.

Migration

Database

In databases, the process of moving data or modifying the database schema (structure) in a controlled, versioned way.

N

2 terms

Normalization

Database

The process of organizing a relational database to reduce data redundancy and improve data integrity by applying a series of normal forms.

NULL

Database

A special marker in SQL indicating the absence of a value — distinct from zero or an empty string.

O

6 terms

Object

Object-Oriented

An instance of a class in OOP that holds state (attributes) and behavior (methods) as defined by its class blueprint.

Object-Oriented Programming

Object-Oriented

OOP — a programming paradigm that organizes software design around objects, which combine data and behavior, guided by principles of encapsulation, inheritance, and polymorphism.

OOP

Object-Oriented

Object-Oriented Programming — a paradigm organizing code around objects that encapsulate state and behavior. See Object-Oriented Programming.

Operating System

Systems

System software that manages computer hardware and software resources, providing common services for application programs.

Overloading

Object-Oriented

Defining multiple methods or operators with the same name but different parameter lists in the same class.

Overriding

Object-Oriented

Redefining a method in a child class that already exists in the parent class, to provide a specific implementation.

P

7 terms

Package

Programming

A namespace that organizes a set of related classes and interfaces in a programming language such as Java or Python.

Pipelining

Systems

A technique where multiple instruction phases overlap in execution, improving CPU throughput by processing several instructions simultaneously.

Pointer

Programming

A variable that stores the memory address of another variable, used for direct memory access in languages like C and C++.

Polymorphism

Object-Oriented

An OOP principle where objects of different classes can be treated as objects of a common superclass, each responding differently to the same method call.

Primary Key

Database

A column or set of columns in a database table that uniquely identifies each row and cannot contain NULL values.

Process

Systems

An instance of a program in execution, with its own memory space, resources, and state managed by the operating system.

Protocol

Network

A set of rules and conventions that govern how data is transmitted and received between devices on a network.

Q

3 terms

Query

Database

A request for data or information from a database, typically written in SQL, that retrieves, inserts, updates, or deletes records.

Query Optimization

Database

The process by which a DBMS automatically chooses the most efficient execution plan for a given query to minimize response time.

Queue

CS Concept

A linear data structure that follows First-In-First-Out (FIFO) order — elements are added at the rear and removed from the front.

R

7 terms

RDBMS

Database

Relational Database Management System — a DBMS that stores data in structured tables with relationships enforced through keys and constraints.

Referential Integrity

Database

A database constraint ensuring that a foreign key value always refers to a valid, existing primary key value in the referenced table.

Repository

DevOps / Cloud

A central location — either a version control system or a package registry — where code, data, or software packages are stored and managed.

Runtime

CS Concept

The period during which a program is executing; also refers to the runtime environment that provides resources for program execution.

Recursion

CS Concept

A programming technique where a function calls itself to solve a smaller instance of the same problem, with a defined base case to stop.

Relational Database

Database

A database that organizes data into tables (relations) with rows and columns, where relationships between tables are defined through keys.

REST

Network

Representational State Transfer — an architectural style for designing networked APIs using HTTP methods (GET, POST, PUT, DELETE) over stateless connections.

S

9 terms

Schema

Database

The formal structure or blueprint of a database, defining its tables, columns, data types, relationships, and constraints.

Software Development Life Cycle

CS Concept

SDLC — the structured process for planning, creating, testing, and deploying software, covering phases from requirements to maintenance.

SQL

Database

Structured Query Language — the standard language for managing and querying relational databases, used for data definition, manipulation, and control.

String

Programming

A data type representing a sequence of characters such as letters, digits, and symbols, used for storing and manipulating text.

Superkey

Database

Any set of attributes that can uniquely identify a row in a database table, including combinations larger than a candidate key.

Server

Network

A computer or software system that provides services, data, or resources to other computers (clients) over a network.

Sorting

Algorithm

The process of arranging data elements in a specified order (ascending or descending) using algorithms like QuickSort, MergeSort, or BubbleSort.

Stack

CS Concept

A linear data structure following Last-In-First-Out (LIFO) order — elements are pushed onto and popped from the same end (top).

Subquery

Database

A query nested inside another SQL query, used to return data that will be used in the outer query's condition or result set.

T

5 terms

TCP/IP

Network

Transmission Control Protocol/Internet Protocol — the core communication protocols governing how data is transmitted across the internet.

Thread

Systems

A lightweight unit of execution within a process that shares the process's memory space and runs concurrently with other threads.

Transaction

Database

A sequence of one or more database operations executed as a single logical unit, either fully committed or fully rolled back.

Trigger

Database

A stored database procedure that automatically executes in response to a specific event such as INSERT, UPDATE, or DELETE on a table.

Two-Phase Locking

Database

2PL — a concurrency control protocol in databases that ensures serializability by acquiring all locks before releasing any.

U

2 terms

UML

CS Concept

Unified Modeling Language — a standardized notation for visualizing software design through diagrams such as class, sequence, and use-case diagrams.

Union

Database

A SQL operator that combines the result sets of two or more SELECT statements, removing duplicate rows by default.

V

3 terms

Variable

Programming

A named storage location in a program that holds a value of a specific data type, which can be changed during execution.

View

Database

A virtual database table defined by a stored SQL query that presents data from one or more underlying tables without storing data itself.

Virtual Machine

Systems

An emulated computer system that runs on physical hardware, providing an isolated environment with its own OS and resources.

W

1 term

Wrapper

Programming

A class or function that encapsulates another component to provide a simplified, adapted, or extended interface to it.

X

1 term

XML

CS Concept

eXtensible Markup Language — a flexible, human-readable format for encoding structured data using custom tags, used in configuration and data exchange.

Z

1 term

Zero-Day Vulnerability

Security

A software security flaw that is unknown to the vendor and has no available patch, making it exploitable from the moment of discovery.

Non-Translatable Technical Glossary · NPTEL CS/SE Course Translation Pipeline · Version 1.0 · All listed terms are DNT (Do-Not-Translate) and must appear unchanged in any translation output. · Categories: OOP · DB · CS-CONCEPT · ALGO · NETWORK · DEVOPS · ML · SECURITY · PROG · SYSTEM